

Reddit Depression Detection using NLP and Transformers

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Introduction

Mental health issues, particularly depression, have become increasingly prevalent in recent years, especially among active social media users. Reddit, as one of the largest online discussion platforms, offers a rich and diverse source of user-generated content that can reflect underlying psychological states. Given the textual nature of Reddit posts and the sensitive context of mental health, Natural Language Processing (NLP) provides a powerful avenue for detecting patterns associated with depression.

The primary objective of this project is to develop an intelligent system capable of automatically classifying Reddit posts into two categories:

Depressed (label = 1) & Non-depressed (label = 0)

Dataset Overview

Source: Kaggle Reddit Depression Dataset

Records: ~2.47 million Reddit posts

Features Available:

- title, body: textual content of the post
- subreddit, created_utc, upvotes, num_comments: metadata
- label: target variable (0 = non-depressed, 1 = depressed)

Before Preprocessing:



Preprocessing Decisions:

To ensure the model learns from the actual linguistic content rather than relying on metadata or shortcut features, we made several key preprocessing decisions:

Removed the subreddit column:

This feature directly indicated the source subreddit (e.g., r/depression, r/teenagers). Including it would make the classification task trivial for the model, as certain subreddits are strongly associated with specific labels. To challenge the model and prevent label leakage, we excluded this column entirely.

Removed the created_utc column:

The post creation timestamp did not add any meaningful semantic information relevant to the user's mental state. Since it introduces noise and has no clear correlation with the label, we discarded it during preprocessing.

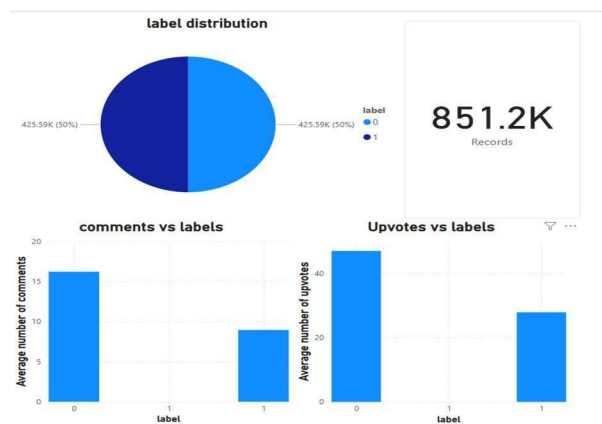
Merged the title and body columns:

Reddit posts often contain valuable context split between the title and the body. To give the model a complete understanding of each post, we combined both fields into a single text input. This enriched the input representation and allowed the model to process the full context in one sequence.

Applied Under-sampling to balance the classes:

The dataset was highly imbalanced, with a significant majority of non-depressed posts (label = 0). To train the model more effectively and avoid bias toward the dominant class, we applied under-sampling. We reduced the number of examples from the majority class to exactly match the number of examples in the minority class. This resulted in a perfectly balanced dataset, allowing the model to learn equally from both classes.

After Preprocessing:



The pie chart illustrates that the dataset is now perfectly balanced after under-sampling each class now constitutes exactly 50% of the dataset.



The chart illustrates clear differences in user engagement between depressed and non-depressed Reddit posts. Specifically, non-depressed posts receive a noticeably higher average number of comments and upvotes compared to their depressed counterparts. This suggests that posts labeled as non-depressed tend to generate more interaction from the community, both in terms of discussion and approval. On the other hand, posts indicating depressive content appear to receive less attention and visibility. These trends highlight how the emotional tone of a post may influence user behavior on the platform, potentially reflecting underlying patterns in social responsiveness and support dynamics.

Models Implemented

DistilBERT is a lightweight and distilled version of the original BERT architecture. It preserves approximately 97% of BERT's performance while reducing the model size by 40% and boosting inference speed by 60%. In this project, DistilBERT was fine-tuned on the balanced Reddit dataset to classify posts based on linguistic indicators of depression. The model demonstrated strong generalization capabilities and fast training, making it suitable for resource-constrained environments.

ELECTRA-small introduces an innovative approach to pretraining language models by replacing the traditional masked language modeling task with a "replaced token detection" objective. Despite its compact size, this variant is highly efficient in terms of both speed and accuracy. When applied in our project, ELECTRA-small effectively captured subtle textual cues related to depressive content, while maintaining computational efficiency.

MiniLM was developed to offer minimal size and high-speed inference while maintaining competitive performance through a compact transformer design. In our implementation, the L6-H384 configuration—featuring 6 layers and 384 hidden dimensions—provided a balanced trade-off between capability and efficiency. Despite its small footprint, MiniLM achieved impressive results in classifying depression-related posts, especially in scenarios requiring deployment on edge devices or limited hardware.

Model Development Workflow

To build a reliable depression classification system for Reddit posts, we followed a structured and reproducible modeling pipeline. This process was carefully crafted to ensure

consistency across different architectures while allowing objective performance comparison.

The dataset split into three distinct subsets:

- Training set (70%) for model learning
- Validation set (15%) for tuning and early stopping
- Test set (15%) for final evaluation

Tokenization and Input Encoding

Given the text-heavy nature of the task, all inputs were tokenized using a suitable tokenizer compatible with the chosen transformer model. To handle large datasets efficiently and avoid memory issues, tokenization was performed in batches. Each sequence was padded and truncated to a fixed length, maintaining uniformity in input shapes.

Custom Dataset Wrapping

To interface with the training framework, we wrapped the tokenized inputs and corresponding labels in a custom PyTorch dataset class. This modular design simplified dataset management and allowed seamless integration with HuggingFace's Trainer API.

Model Initialization and Training Configuration

A pre-trained transformer model was loaded and configured for binary sequence classification. The model was transferred to GPU (if available) to accelerate training.

Training arguments:

- Epoch count
- Batch size
- Learning rate
- Weight decay
- Evaluation strategy
- Early stopping to prevent overfitting

These parameters were chosen to strike a balance between performance and efficiency

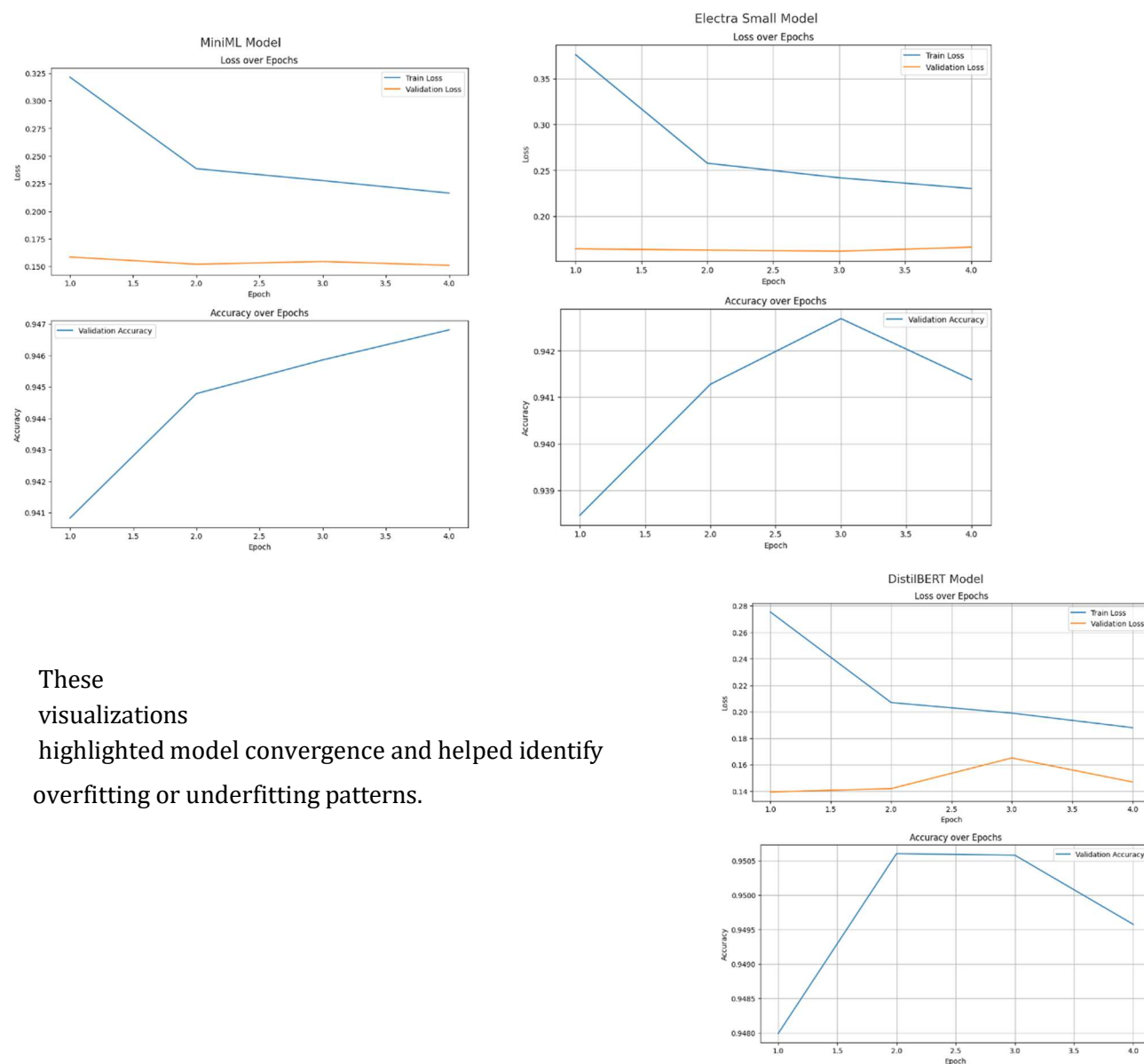
Training and Evaluation

The model was then fine-tuned on the training data using the HuggingFace Trainer API. During each epoch, the model was evaluated on the validation set, with the best-performing checkpoint automatically saved based on validation accuracy.

Upon training completion, the model was evaluated on the unseen test set, where performance metrics such as accuracy, precision, recall, and F1-score were computed to assess its generalization ability.

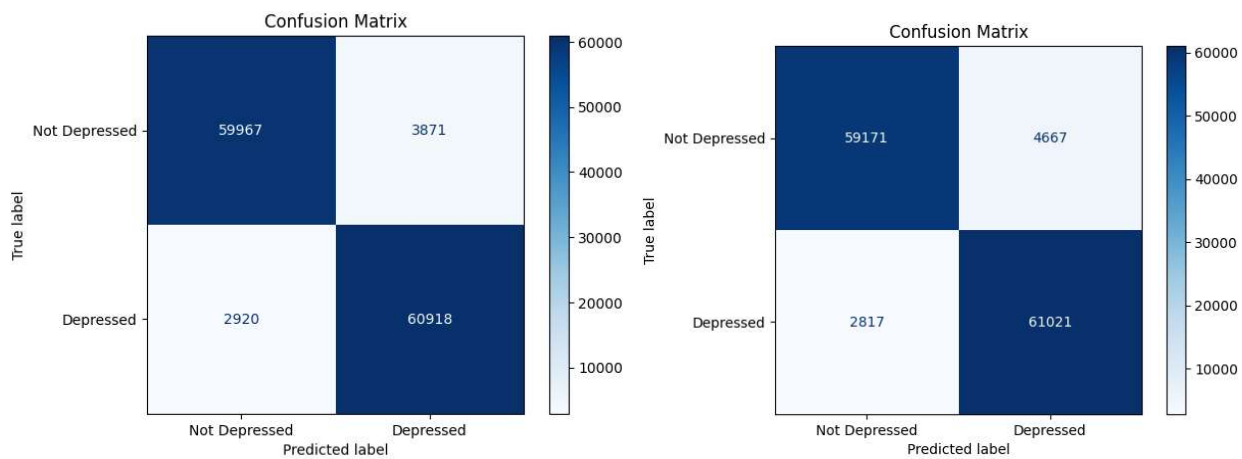
Performance Visualization

Training vs validation loss & Accuracy curves



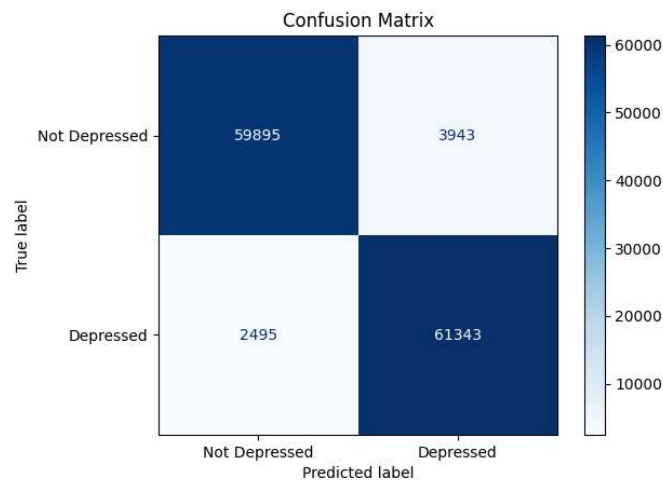
These visualizations highlighted model convergence and helped identify overfitting or underfitting patterns.

confusion matrices



MiniLM

ElectraSmall



DistilBert

classification report

Model	Precision	Recall	F1-Score	Accuracy
MiniLM	0.95	0.95	0.95	0.95
DistilBERT	0.95	0.95	0.95	0.95
ELECTRA-small	0.94	0.94	0.94	0.94

This diagnostic step revealed important nuances in model behavior and guided future improvements.

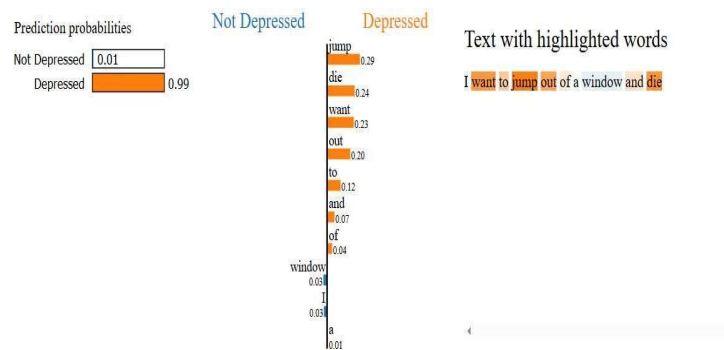
Attention-Based Interpretability

We applied **LIME** (Local Interpretable Model-Agnostic Explanations) to interpret the predictions generated by our models.

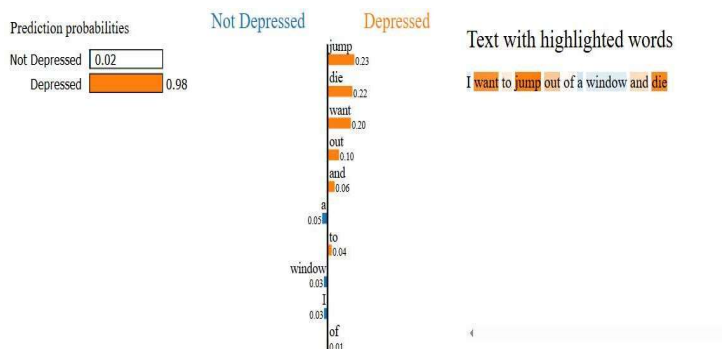
LIME identifies the most influential tokens contributing to each classification decision by approximating the model's behavior locally using an interpretable surrogate model.

The explanations were presented through color-coded highlights, allowing us to visualize which words had the greatest impact on the output.

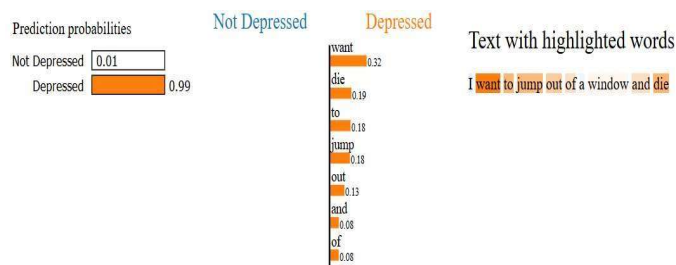
This interpretability analysis enabled us to verify whether the model's predictions aligned with meaningful linguistic patterns commonly associated with depressive or non-depressive language.



ELECTRA-small



Distilbert



MiniLM-

These insights demonstrated which words the model focused on when predicting a user's mental state, providing a valuable interpretability layer to the system.

Comparative Conclusion

The three transformer-based models—**DistilBERT**, **ELECTRA-small**, and **MiniLM**—demonstrated strong performance in detecting depressive language in Reddit posts. Despite having different architectures and sizes, they all achieved comparable F1-scores (≈ 0.94 – 0.95).

- **DistilBERT** (~66M parameters) delivered high accuracy with fast inference, suitable for general-purpose applications.
- **ELECTRA-small** (~14M parameters) was efficient and lightweight, capable of identifying subtle linguistic cues with minimal resources.
- **MiniLM** (~22M parameters) provided the **best trade-off** between performance, size, and speed—making it the most practical choice for both accuracy and deployment on limited hardware.

In conclusion, **MiniLM** stands out as the most balanced model, combining compact architecture with strong classification capabilities.

Model Saving and Deployment

Finally, the trained model and tokenizer were saved locally, ensuring they could be reloaded for inference or deployment in any environment. This step paved the way for integrating the model into a real-time GUI interface.

Independent Model Implementation

Each transformer model was implemented and trained in a dedicated notebook, following this exact pipeline. This design ensured clear experiment tracking, reproducibility, and clean separation of results, enabling fair and structured comparisons

GUI Demonstration (Gradio)

To enable easy and interactive testing, we developed a **Gradio-powered GUI** that allows users to input any Reddit-style sentence and receive real-time predictions from the three models. As shown in the image, the interface includes:

- A text box for sentence input
- A submit button to generate predictions
- Side-by-side results from **DistilBERT**, **ELECTRA**, and **MiniLM**, each showing the predicted label and confidence percentage

Depression Detection using 3 Models

This tool uses 3 different transformer models to predict if the sentence indicates depression or not.

Enter a sentence

I want to jump out of a window and die

Clear

Submit

DistilBERT Prediction

Depressed (98.64%)

Electra Prediction

Depressed (99.33%)

MiniLM Prediction

Depressed (98.22%)

Flag